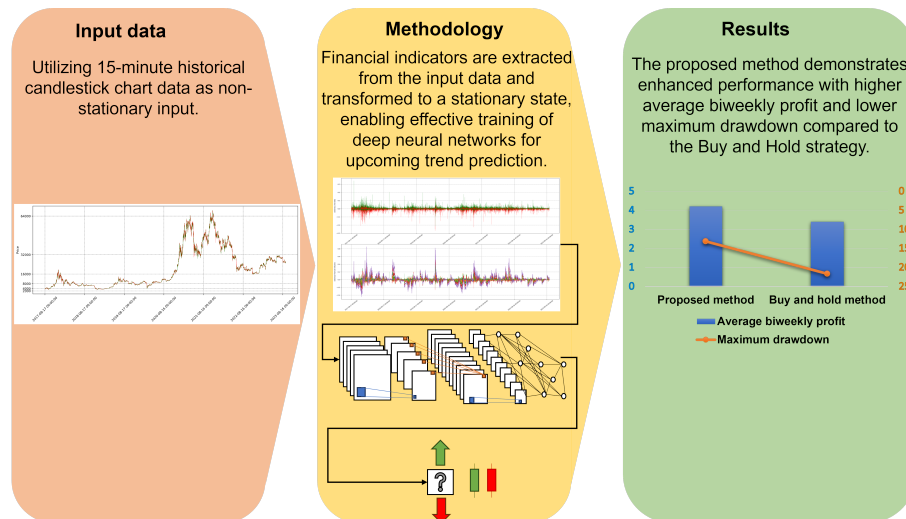


Graphical Abstract

Deep High-Frequency Cryptocurrency Trend Detection: An Approach For Data Stationarity

Ali Asareh Nejad, Ahmad Kalhor, Reshad Hosseini, Babak Nadjar Araabi



Highlights

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- Advances the quest for optimal models in cryptocurrency prediction by evaluating multiple deep learning methods with a comprehensive set of evaluation metrics.
- Addresses the under-explored domain of intraday trading with a focus on 15-minute candlestick data using the CNN network, offering new insights into high-frequency trading dynamics.
- Focuses exclusively on intrinsic features and incorporates novel technical indicators tailored to the cryptocurrency market's volatility.
- Introduces a methodological innovation that ensures data stationarity and systematically evaluates the impact of the relative positioning of financial features, enhancing prediction accuracy.
- Uses a novel weighted loss function, balancing accuracy with resilience to outliers.

Deep High-Frequency Cryptocurrency Trend Detection: An Approach For Data Stationarity

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Abstract

Predicting Bitcoin's market trend poses a significant challenge for researchers, investors, and traders alike. Although the relevance of daily time frames in cryptocurrency prediction is widely acknowledged, the potential of shorter intraday periods remains underexplored. This study examines the use of 15-minute candlestick data through Convolutional Neural Networks (CNN) for deep learning-based trend prediction in the cryptocurrency market. We introduce a novel data preprocessing method that ensures stationarity while preserving essential relationships between indicators and candlestick data. Upon evaluating various neural network architectures, our proposed CNN model demonstrates superior performance, achieving an accuracy of 53.955%. It notably enhances average biweekly returns to 4.182% and decreases maximum drawdown, surpassing the Buy and Hold strategy. Our research addresses critical issues, including the impact of data transformation for stationarity, the implementation of innovative loss functions to handle strategic misclassifications, and the integration of new indicators to boost the model's predictive power. The results highlight the model's enhanced ability to predict trends, emphasizing the importance of specific data preprocessing and model configuration.

Keywords: Stationary Transformation in Time Series, Technical Indicators for Trading Algorithms, Loss Function Optimization in Machine Learning, Deep Learning for Cryptocurrency Prediction, High-Frequency Cryptocurrency Trading, Intraday Market Trend Analysis

1. Introduction

Cryptocurrencies, led by Bitcoin Wright (2008), have revolutionized the financial landscape by offering an alternative to traditional, centralized financial

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systems Motamed and Bahrak (2019). Since Bitcoin's introduction, it has become a distinct asset class, yet its market trend prediction remains less explored compared to traditional stock markets Brière et al. (2015), identifying a crucial research gap McNally et al. (2018). The shift towards fundamental over technical analysis methodologies, along with the impact of widespread financial news BROCK et al. (1992), complicates the prediction landscape. Opinions vary widely on the predictability of Bitcoin's price, from it being perceived as a random walk Stavroyiannis et al. (2019) to exhibiting near-random walk behaviors Pintelas et al. (2020), underscoring the need for further investigation.

Research into forecasting financial time series has encompassed classical statistical methods, machine learning strategies, artificial neural networks, and hybrid approaches. Classic statistical models like ARIMA and GARCH have demonstrated utility in historical price analysis Amjad and Shah (2017), and uncovering market anomalies Shanaev et al. (2023). However, their reliance on rigid assumptions often limits real-world applicability Khedr et al. (2021). Traditional Machine learning methods including SVM Mahdi et al. (2021), logistic regression Akyildirim et al. (2021), K-nearest neighbors Amjad and Shah (2017), decision trees Jaquart et al. (2021), and boosted trees Oyedele et al. (2023) have shown promise, with SVM frequently outperforming others Nikou et al. (2019); Akyildirim et al. (2021). However, they often struggle to fully capture cryptocurrency markets' temporal depth and volatility Livieris et al. (2020a); Shen et al. (2020). Artificial neural networks, especially LSTM networks, have demonstrated superior accuracy Murray et al. (2023) over statistical models and machine learning techniques but face challenges like overfitting Kim and Sung (2022) and optimizing architectures Seabe et al. (2023).

The increasing reliance on trend prediction for financial decision-making has shifted focus towards Machine Learning (ML) algorithms, recognized for their ability to uncover nonlinear data relationships beyond conventional statistical models' capabilities Vidal and Kristjanpoller (2020). This trend has particularly boosted the prominence of Algorithmic and High-Frequency Trading (HFT), which rely on exploiting minor price fluctuations over short periods, a strategy made increasingly relevant by the intraday volatility characteristic of cryptocurrency markets Huang et al. (2019a).

As cryptocurrencies become more established in the financial sector Motamed and Bahrak (2019), the interest in leveraging Neural Networks like Multi Layer Perceptron (MLP), CNN, and hybrid CNN-LSTM for the prediction of short-term price movements grows Alonso-Monsalve et al. (2020). Despite significant advancements in deep neural network architectures for analyzing both stock Chong et al. (2017) and cryptocurrency markets Lahmiri and Bekiros (2019), the high market volatility Shintate and Pichl (2019) and the issue of data non-stationarity Chang et al. (2023); Shang et al. (2023) present ongoing challenges. Current methodologies have made strides in addressing these complexities; however, they frequently fall short in considering the crucial role of the relative positioning of input features which contains valuable information about the dynamic of the market. This oversight can lead to inaccuracies in predictions, potentially resulting in considerable financial losses.

This study introduces a novel technique for data stationarity, preserving essential information, and incorporates new indicators based on traders' insights. A custom loss function is also developed to enhance prediction accuracy. Furthermore, it evaluates multiple deep learning models for cryptocurrency prediction, using a comprehensive set of evaluation metrics. Our framework for predicting cryptocurrency trends significantly improves accuracy and profitability.

The remainder of this paper is organized as follows: Section 2 reviews the relevant literature, Section 3 describes the research methodology, and Section 4 presents the results. Section 5 consolidates the findings, discusses their implications, and proposes directions for future research, culminating in a summary of the study's principal contributions.

2. Related work

Deep learning has significantly impacted time series forecasting by developing highly effective models Livieris (2023). The adaptability of deep learning architectures has facilitated unparalleled progress in predicting complex financial trends. Comparative studies highlight deep learning's edge in market predictions. Nabipour et al. (2020) contrasts MLP, Recurrent neural network (RNN), and Long Short Term Memory neural network (LSTM) with conventional algorithms like Decision Trees and eXtreme Gradient Boosting (XGBoost), demonstrating deep learning's superiority in stock trend detection. Similarly, Oyedele et al. (2023) shows that deep learning models, particularly MLP, Gated Recurrent Units (GRU), and CNN, outperform ensemble methods including XGBoost and Adaptive Boosting (AdaBoost) in cryptocurrency price forecasting. Murray et al. (2023) further broadened the comparative analysis to include statistical models, machine learning, and deep learning, consistently finding LSTMs to outperform other models for price prediction across various cryptocurrencies. Conversely, Cavalli and Amoretti (2021) demonstrated that their 1D CNN model, trained on multivariate datasets, predicts bitcoin trends with greater accuracy than LSTM models. Additionally, a review by Shah et al. (2022) highlighted LSTM and its hybrids' success in stock price prediction, whereas CNNs and their hybrids were more proficient in trend prediction. While progress has been notable, the search for optimal models continues. This research contributes to the field by evaluating a variety of deep learning methods based on summary statistics of multiple evaluation metrics, an approach not extensively explored in the existing literature.

For readers interested in a deeper exploration of financial time series forecasting through deep learning, Sezer et al. (2020) provide an extensive overview. Nazareth and Ramana Reddy (2023) review machine learning methodologies specifically tailored to financial markets, offering insights into current trends and applications. Khedr et al. (2021) provides a survey on traditional statistical techniques and machine learning approaches for cryptocurrency predictions. Additionally, Shah et al. (2022) present a comprehensive analysis of hybrid deep learning techniques applied to stock predictions. Together, these works lay the foundation for understanding the complexities of financial market predictions,

upon which this study builds by evaluating the effectiveness of various deep learning methods for cryptocurrency trend prediction.

Intraday technical trading in financial markets has attracted considerable research attention, particularly regarding the dynamics of short-term price movements. Nakano et al. (2018) investigated intraday trading of Bitcoin using artificial neural networks and demonstrated significant performance improvements over the buy-and-hold strategy, especially during volatile periods. Similarly, Chen et al. (2020) recognized Bitcoin's volatility and employed two different modeling techniques for datasets with differing structures and dimensional features to predict Bitcoin prices at different frequencies, achieving notable accuracy for both daily and 5-minute intervals. Furthermore, Khurana et al. (2023) developed models for AI-based trading bots capable of predicting various components of Bitcoin candlesticks across different intraday time frames, thus showcasing the potential of machine learning in high frequency cryptocurrency predicting. Although significant strides have been made, the realm of intraday trading for cryptocurrencies remains underexplored. This study aims to bridge this gap by analyzing 15-minute chart data with a focus on the CNN network, a methodology not extensively examined in the existing literature. Through this approach, we seek to contribute new insights into the effectiveness of deep learning methods for intraday cryptocurrency trend prediction.

The debate on optimal input features for cryptocurrency prediction continues, with research exploring both exogenous features, such as economic variables and social media sentiment analysis Georgoula et al. (2015); Mittal et al. (2019), and intrinsic features. Bâra and Oprea (2024) forecast Bitcoin closing prices over the next 7 days using an Ensemble Learning Method that combines historical Bitcoin data, volatility indicators, and binary trend prediction. This method leverages the impact of recent history on current prices and integrates 38 technical volatility indicators and market trend analyses to improve predictive accuracy. Similarly, Huang et al. (2019b) investigates Bitcoin's return predictability through a classification tree-based model using 124 technical indicators. Despite the potential advantages of exogenous features, their application is hindered by challenges such as the scarcity of high-frequency data, lengthy data collection times, high access costs, and significant computational demands, which could impede real-time prediction frameworks. This study exclusively focuses on intrinsic features to mitigate these challenges and distinguishes itself by incorporating novel technical indicators, specifically selected for their relevance to the cryptocurrency market's volatility.

The issue of non-stationarity in financial data has been a focal point in multiple studies. For instance, Khosravi and Ghazani (2023) introduced a novel data pre-processing and feature selection method utilizing machine learning techniques (RF, XGBoost, LSTM) applied to Bitcoin return series. Their approach resulted in superior performance and reduced generalization error. Livieris et al. (2020b) introduced a framework that enhances deep learning models for time-series forecasting by ensuring stationarity through transformations such as first differences validated by the Augmented Dickey-Fuller test. This method improves model predictions by eliminating auto-correlation in errors, thereby

boosting the accuracy and reliability of forecasts. Livieris (2023) developed a strategy based on the iterative application of a series of transformations until the time-series becomes stationary, with their experiments confirming the method's effectiveness in financial and commodity markets. Despite these advancements, the literature has not thoroughly investigated how the relative positioning of financial features, such as the placement of the 100 candle simple moving average relative to the current candle's closing value, on prediction accuracy remains underexplored. Addressing this gap, our study introduces an approach that ensures data stationarity and systematically evaluates the impact of financial features' relative positions, aiming to deepen our understanding of market dynamics and improve prediction accuracy in previously uncharted areas.

Recent literature underscores the critical role of reward/loss function selection in enhancing cryptocurrency price prediction accuracy. Utilizing the Deep Q-Network (DQN) framework, Muminov et al. (2024) introduced a complex reward function that augments prediction precision by incorporating confidence scaling and time-based discounting. Similarly, Koo and Kim (2024) targeted improvements in Bitcoin's price tail regions with a novel loss function that variably penalizes based on label distribution specifics. Even with recent improvements, the exploration for optimal loss functions in cryptocurrency trend classification continues to be a promising avenue for further investigation. In response, our study introduces a weighted loss function combining mean squared error (MSE) and mean absolute error (MAE), a method rarely employed in this context, striving for a balance between accuracy and resilience to outliers and advancing the domain of cryptocurrency price prediction.

Notably, despite significant progress, the field faces ongoing challenges, particularly in choosing the optimal models, making the features stationary while maintaining crucial information of the market data and incorporating effective loss functions in volatile markets. Addressing these issues is key to advancing our understanding and capabilities in cryptocurrency price prediction.

3. Methodology

This section explains the approach and techniques employed in this study for predicting cryptocurrency market trends. Our framework integrates several novel techniques, each aiming to enhance the accuracy of predictions. The flowchart of the proposed framework is depicted in Figure 1, highlighting the innovative elements of the model development process.

3.1. Data Stationary Transformation and Enhanced Analysis

The development of accurate prediction models faces many challenges, with data non-stationarity being a significant concern. This issue, marked by changing statistical properties of data over time, can bias and reduce the reliability of predictions. To address this critical challenge, we introduce a novel method involving the calculation of percentage deviation from each candlestick's opening price. This approach provides a stationary representation of the indicator,

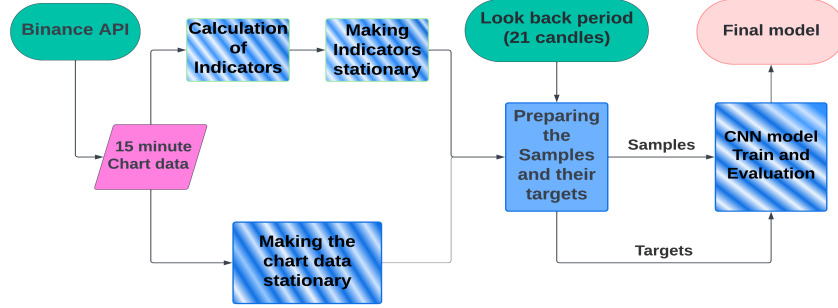


Figure 1: Flowchart illustrating the proposed CNN model development for cryptocurrency trend prediction. Innovations, distinguished by striped boxes, encompass the computation of novel indicators, specialized methods for data stationarity, and the adoption of a new loss function for model training. Detailed descriptions of these advancements are provided in the Methodology section.

retaining its essential characteristics and the position of this indicator with respect to other indicators and candle stick chart. The stationary value, denoted as indicator_{s_i} , is calculated as follows:

$$\text{indicator}_{s_i} = \frac{(\text{indicator}_{ns_i} - \text{open}_{c_i})}{\text{open}_{c_i}} \quad (1)$$

where indicator_{ns_i} represents the non-stationary indicator value for the i -th candlestick, and open_{c_i} denotes the opening price for that candlestick. Notably, for non-stationary volume-derived indicators, we achieve stationarity through a logarithmic transformation.

The advantages of addressing data stationarity are readily apparent. By ensuring stationarity we prevent the model from being biased and, the model could learn the relation of features and the response variable more effectively. Consequently, this leads to a significant enhancement in the reliability, efficacy, and accuracy of our cryptocurrency price prediction framework. The transformative impact of this approach is illustrated in figure 2.

Beyond the commonly utilized indicators in literature and those available in the Python Ta library, we have implemented and incorporated a suite of established technical indicators, favored by traders across various financial markets and tailored to the unique dynamics and volatility of the cryptocurrency market. These include the Closest Support/Resistance, the Golden Resistance/Support indicator, and the Dynamic Resistance/Support indicators. For accessibility, the Python implementation of these indicators, the data used in this study, and our methodology are made available on the corresponding author's GitHub page¹.

¹<https://github.com/AliAssareh/stationary-trend-prediction>



Figure 2: Comparative visualization of BTCUSDT market data and indicators before and after applying the proposed data stationarity method. Part (a) displays the original 15-minute candlestick chart with four derived indicators, showcasing the market dynamics without stationarity adjustments. Part (b) illustrates the corresponding period after applying the stationarity transformation, demonstrating the method’s impact on the data’s temporal characteristics. The graphical representation emphasizes the effectiveness of the proposed methodology in preparing the market data and indicators for subsequent trend prediction analysis.

3.2. Predictor Network: Architecture and Training

Among the objectives of this study was to identify the most effective model within our proposed framework for cryptocurrency market trend prediction. A comprehensive evaluation was undertaken across various models, including MLP, CNN, RNN, GRU, LSTM, and hybrid CNN-LSTM, each tested with a range of hyper-parameters. Models that demonstrated superior performance on validation sets were selected. The specific configurations of these leading models are available on the corresponding author’s GitHub page. The selected configuration for each model was instantiated from 32 distinct initial states, trained as outlined below, and evaluated on the test set to generate a random sample of results for model comparison. These random samples underwent an outlier removal process using the Inter Quartile Range (IQR) method, where the cutoff for outliers is set at 1.5 times the IQR from the 25th and 75th percentiles. The motivation behind this was to eliminate the results from models that were not properly trained, which can occur for neural networks if the initial point is not appropriate.

Our data obtained from Binance API, covering 15-minute BTC/USDT candlestick charts from August 28, 2017, to January 03, 2024, was partitioned into training (75%), validation (15%), and test (10%) segments, each encompassing bullish and bearish market phases. The training phase employed various techniques to refine model performance, including the Adam optimizer for dynamic

learning rate adjustments, L2 regularization, and early stopping to mitigate overfitting. Additionally, sample weighting was applied, inversely proportional to class frequency, to address dataset imbalance.

For model development we utilized PyTorch, which provides a potent and adaptable environment for deep neural network training. A look-back period of 21 candles, was determined through exhaustive analysis, to balance historical data integration with model complexity. To enhance classification, a loss function combining weighted averages of Mean squared error (MES) and mean absolute error (MAE) penalties, unprecedented in cryptocurrency trend prediction to our knowledge, was adopted. This function aims to reduce false predictions while tolerating minor misclassifications. The efficacy of this approach is substantiated through empirical testing, further detailed in Section 4.

3.3. Trading Strategy

The primary focus of this study was to evaluate the efficacy of the proposed predictive framework, rather than to develop an exhaustive trading system, which would necessitate addressing complexities such as time management, liquidity constraints, and transaction costs. However, to illustrate the practical application of our framework, a trading strategy predicated on model predictions was considered, and model outcomes evaluated under the assumptions of this strategy.

In the deployed strategy, receipt of a buy signal, indicative of an anticipated price increase, prompts a 100\$ purchase of Bitcoin at the closing price of the relevant 15-minute candle. This position is maintained until the closure of the subsequent candle, at which point the acquired Bitcoin quantity is sold at the new candle's closing price. Key assumptions include immediate execution of buy and sell orders upon candle closure and sufficient market liquidity to fill these orders, aligning with prevalent assumptions in this field's literature.

The strategy's risk and performance evaluation is conducted using critical metrics such as win rate, profitability, and maximum drawdown, detailed in the 4 section. Integration of the proposed framework into this systematic trading strategy aims at demonstrating its effectiveness in identifying profitable trading opportunities.

4. Results and discussion

This section presents the results and evaluates the predictive models developed in this study. It details the models' accuracy, assesses their profitability within the applied trading strategy, and compares our methodology to established methods in the field. Furthermore, it addresses the study's core questions and provides a comprehensive discussion of the findings in relation to the research questions posed.

The primary objective of this research is to determine if a transformation exists that could render indicator data stationary while preserving their relative positions to each other and to candlestick chart data, and crucially, if such

a transformation could enhance trend prediction accuracy. Additionally, we explore whether it is possible to design a loss function that tolerates certain misclassifications to boost the model’s overall performance in cryptocurrency trend prediction. We also investigate the potential of integrating new indicators to improve model predictions. Finally, we seek to identify which model would excel within this framework.

The first experiment compares the performance of various neural network models within our predictive framework to identify the model with the highest efficacy for trend prediction under the specified configuration. The accuracy of these models is presented in figure 3. Additionally, we present performance metrics for these models, including accuracy, area under the curve (AUC) of the receiver operating characteristics curve, and their performance under the proposed trading strategy. This includes the average biweekly return (ABR) and the maximum drawdown (MDD) which is the largest observed loss of capital during the test period, expressed as a percentage. These metrics, alongside those for the Buy and Hold (B&H) strategy, are detailed in Table 1. Furthermore, Figure 4 displays the biweekly profit details of the models.

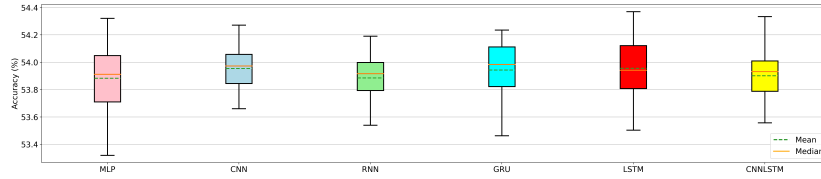


Figure 3: Box plots illustrating the distribution of accuracy for various neural network models. Each box represents a different model, detailing the spread of accuracy scores.

Table 1: Performance Comparison of Neural Network Models and Buy and Hold Strategy.

Method	Accuracy (%)	AUC	Average biweekly Return (%)	MDD (%)
MLP	53.883 ± 0.096	0.556 ± 0.001	3.620 ± 0.151	11.817 ± 0.582
CNN	53.955 ± 0.055	0.557 ± 0.001	4.182 ± 0.123	13.176 ± 0.681
RNN	53.886 ± 0.056	0.556 ± 0.001	3.375 ± 0.150	12.502 ± 0.971
GRU	53.927 ± 0.079	0.557 ± 0.001	3.394 ± 0.153	12.767 ± 0.640
LSTM	53.956 ± 0.079	0.557 ± 0.001	3.362 ± 0.180	13.033 ± 0.434
CNNLSTM	53.902 ± 0.062	0.556 ± 0.001	3.557 ± 0.148	13.303 ± 0.631
Buy and Hold	-	-	3.450	21.705

Upon analyzing the performance of the evaluated models, the CNN model outperformed the others with an accuracy of $53.955 \pm 0.055\%$ and exhibited a superior average biweekly return of $4.182 \pm 0.123\%$. This suggests its enhanced ability to leverage spatial-temporal features for market prediction. Although the

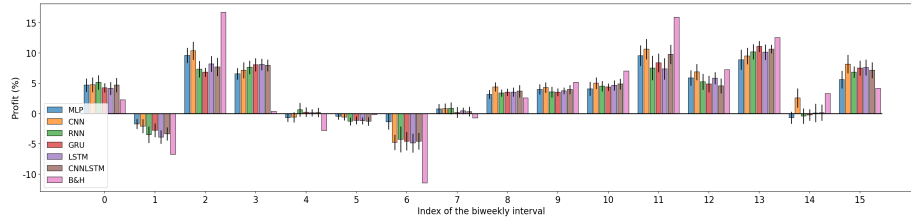


Figure 4: Bar chart illustrating the biweekly profit performance for a range of neural network models, including MLP, CNN, RNN, GRU, LSTM, CNNLSTM, against the traditional Buy and Hold (B&H) strategy, measured over successive two-week periods.

LSTM model showed a marginally higher average accuracy ($53.956 \pm 0.079\%$), it was not selected due to its considerably larger confidence interval, indicating less reliability in its performance metrics. All models displayed closely matched AUC values around 0.556, suggesting similar capabilities in trend discrimination. Notably, the maximum drawdown for the CNN model ($13.176 \pm 0.681\%$) was only slightly higher than those of other models, indicating a balanced risk profile in pursuit of higher returns. This nuanced difference in risk underscores the effectiveness of neural network models, including the CNN, in achieving competitive returns with manageable risk levels. In contrast, the traditional Buy and Hold strategy exhibited a significantly higher maximum drawdown of 21.705% without corresponding return benefits, emphasizing the neural network models' advantage in offering dynamic and risk-adjusted investment strategies for cryptocurrency trading without markedly increasing exposure.

For the CNN model, which emerged as the best performer, key performance metrics, such as accuracy, true positive rate, and Area Under Curve (AUC), are comprehensively detailed in Table 2. Trading performance metrics, including win rate, average biweekly return, and maximum drawdown, are systematically outlined in Table 3.

Table 2: Quantitative Performance Metrics of the CNN Model Across Datasets

Dataset	Accuracy (%)	True Positive Rate (%)	AUC
Train	54.786 ± 0.063	55.051 ± 0.139	0.569 ± 0.001
Validation	54.635 ± 0.067	54.735 ± 0.193	0.565 ± 0.001
Test	53.955 ± 0.055	54.174 ± 0.166	0.557 ± 0.001

To address the first three research questions and demonstrate the effectiveness of the innovations introduced in this study, we conducted a series of experimental analyses. These experiments were designed to compare the impact of each innovation by either excluding it or substituting it with a conventional approach documented in the literature. The specific experiments conducted include:

1. **Log-Difference Stationary** - Substituting the proposed method for

Table 3: Comparative Performance Metrics of the CNN Model and Buy & Hold Strategy Across Datasets.

Strategy	Dataset	Win Rate (%)	Average Biweekly Return (%)	MDD (%)
Proposed (CNN)	Train	54.786 ± 0.063	71.843 ± 2.276	14.854 ± 1.291
Proposed (CNN)	Validation	54.635 ± 0.067	3.999 ± 0.364	42.850 ± 1.871
Proposed (CNN)	Test	53.955 ± 0.055	4.182 ± 0.123	13.176 ± 0.681
Buy & Hold	Train	-	2.718	84.058
Buy & Hold	Validation	-	0.360	51.281
Buy & Hold	Test	-	3.450	21.705

achieving stationarity with the traditional approach of successive differences of the natural logarithm of price Stavroyiannis et al. (2019).

2. **Excluded** - Omitting the indicators presented in this paper.
3. **Loss: MSE** - Replacing the proposed loss function with the Mean Squared Error.

The accuracy of the chosen CNN model, under each experimental condition outlined previously, is illustrated in the box plot presented in Figure 5. Additionally, Table 4 provides a detailed comparison, highlighting the differences in key metrics such as accuracy, AUC, average biweekly return, and maximum drawdown between the proposed framework and the conditions specified in the experiments.

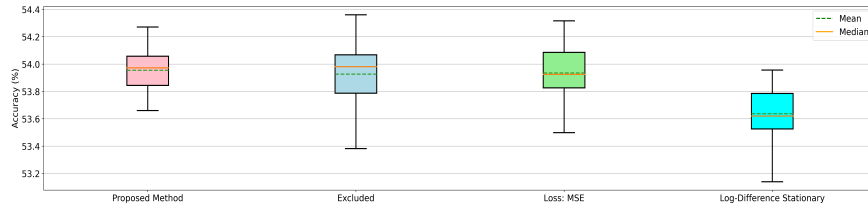


Figure 5: Box plots displaying the distribution of accuracy for the proposed CNN model across different experimental conditions: the standard log-difference stationarity, exclusion of new indicators, and utilization of Mean Squared Error (MSE) as the loss function.

The results, presented in Table 4, highlight the positive impact of our innovations on the performance of CNN model for cryptocurrency trend prediction. Our approach, with its distinct handling of non-stationary in the data, new indicators, and innovative loss function, achieved slightly higher accuracy ($53.955 \pm 0.055\%$) with smallest confidence interval and the highest average biweekly return ($4.182 \pm 0.123\%$) among the tested configurations. Notably, replacing our method of achieving data stationarity with the log-difference technique significantly reduced both accuracy ($53.637 \pm 0.067\%$) and average biweekly return ($3.176 \pm 0.174\%$), demonstrating the superiority of our method in making data stationary while preserving informative relation of the input

Table 4: Comparative performance metrics of the proposed cryptocurrency trend prediction method under various experimental conditions. This table quantitatively illustrates the effects of excluding certain innovations, using Mean Squared Error (MSE) as the loss function, and applying Log-Difference for stationarity.

Method	Accuracy (%)	AUC	Average Biweekly Return (%)	MDD (%)
Proposed method	53.955 ± 0.055	0.557 ± 0.001	4.182 ± 0.123	13.176 ± 0.681
Excluded	53.927 ± 0.081	0.557 ± 0.001	4.124 ± 0.122	13.225 ± 0.657
Loss: MSE	53.935 ± 0.073	0.557 ± 0.001	4.048 ± 0.155	12.703 ± 0.638
Log-Difference Stationary	53.637 ± 0.067	0.552 ± 0.000	3.176 ± 0.174	11.257 ± 1.252

features. Similarly, excluding our newly introduced indicators and using Mean Squared Error (MSE) instead of our novel loss function resulted in slight decreases in performance metrics. However, these reductions were less significant than those observed with the change in the stationarity approach.

Interestingly, the inclusion of new indicators primarily influenced the model's prediction reliability, as evidenced by a narrower confidence interval (improving from ± 0.081 to ± 0.055). This enhancement in the confidence interval underscores the importance of integrating novel indicators, which, while not drastically altering accuracy, significantly bolster the robustness and consistency of trend predictions. Consequently, these experimental findings underscore the effectiveness of our proposed methodology in cryptocurrency trend prediction, highlighting how our novel approach to achieving stationarity, along with the integration of new indicators, contributes to greater prediction confidence and improved model performance. Such advancements underscore the proposed method's potential to exploit profitable trading opportunities within the highly volatile cryptocurrency market effectively.

5. Conclusion and Future Directions

This study presents a comprehensive examination aimed at advancing the prediction of cryptocurrency market trends. The CNN model's notable performance, with an average accuracy of 53.955%, indicates a strong capability to accurately assess Bitcoin market dynamics and confirms the practicality of its application in trading scenarios. When compared to conventional Buy and Hold strategies, which yielded a biweekly profit of 3.450% and a maximum drawdown of 21.705%, our proposed method demonstrated a significantly improved biweekly profit of 4.182% and a reduced maximum drawdown of 13.176%. These results exemplify the potential of our methodological innovations to outperform standard trading strategies by providing both higher profitability and reduced risk.

Through rigorous comparative analysis and experimental validation, this research contributes a robust framework to the body of existing financial modeling literature. The integration of novel indicators and the optimization of the loss function without compromising data integrity or predictive power substantiates the efficacy of our approach.

Despite promising results, the study identifies potential areas for refinement, such as target definition, stop-loss criteria, and data quality assurance. Future research should focus on innovating in target definition to accommodate market volatility, formulating stop-loss strategies to mitigate risks, and exploring the predictive capacity across different temporal resolutions. These avenues seek to address the current study's limitations and to enrich the domain of cryptocurrency price prediction with more precise and robust forecasting models.

Ultimately, this study represents a significant stride in the domain of cryptocurrency trend prediction. It advocates for ongoing innovation and research, aiming to fine-tune predictive accuracy, and integrate comprehensive risk management strategies. The work emphasizes the vital link between machine learning and financial forecasting, setting the stage for future progress in achieving reliable and accurate market trend predictions.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used the Openai GPT-4 service in order to improve the structure of sentences. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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